Potential Outcomes and Causality: Treatment Effects

Joan Llull

I. Introduction

According to *The New Palgrave: A Dictionary of Economics*, econometrics is the branch of economics that aims to give empirical content to economic relations. We use econometrics to this end in three different ways:

- ***Associational inference***: to quantitatively describe or summarize the co-movement of different economic variables. These are widely used to establish "facts" that motivate economic theory, but do not aim to provide, by themselves, evidence about the mechanisms underlying the relation between the variables. They have regained popularity with the new availability of very large datasets ("big data").
- ***Forecasting***: to predict one variable based on the movements of other variables, without aiming, once again, to describe the economic mechanisms of interest. For example, the presence of many people carrying an umbrella helps predict that it is likely to rain, but no one would argue that it rains *because* many people carry umbrellas.
- ***Causal inference***: the process of drawing a conclusion about a cause-effect connection between economic variables. For example, we may be interested in whether hospital stays improve the health of inpatients. The association between hospital stays and health is typically negative, since inpatients are usually sicker than average. However, is this because hospitals worsen health (e.g. exposure to other illnesses), or simply because unhealthy individuals go to hospitals? Causal inference aims to answer *what if* questions. Hence, we want to know whether *the same inpatient* would be more or less healthy if she were sent home versus kept at the hospital, netting out the fact that individuals who go to hospitals are unhealthier to begin with.

In this course we focus on the latter, which is a powerful tool for the *ex-post* evaluation of public policies. The central objective is to establish credible causal links between policy interventions and economic outcomes.

The evaluation of public (and private) policies is important for efficiency and, ultimately, for improving welfare. There is a vast literature in economics—mostly in public economics, but also in development economics and labor economics—devoted to the evaluation of different programs. Examples include training programs, welfare programs, wage subsidies, minimum wage laws, taxation, Medicaid and other

health policies, school policies and feeding programs, microcredit, and a variety of other forms of development assistance. These analyses aim to quantify the causal effects of such policies on economic outcomes and, ultimately, on welfare.

The classic approach to quantitative policy evaluation is the ***structural approach***. This approach specifies a class of theory-based models of individual choice, selects the model within the class that best fits the data, and uses it to evaluate policies through simulation. Its main advantages are that it allows both *ex-ante* and *ex-post* policy evaluation, and that it permits the evaluation of counterfactual policy variations without changing the structure of the model or re-estimating it (out-of-sample simulation).

The main critique of this approach is that it relies on a host of untestable functional form and behavioral assumptions, whose implications for the results are often unclear. These assumptions give researchers substantial discretion, and the complexity of structural models can undermine transparency and replicability. Some authors have argued that this approach places too much emphasis on external validity at the expense of a more basic concern for internal validity.

Over the last two decades, the ***treatment effects approach*** has established itself as an important alternative to the structural approach, introducing a different language, different priorities, and different empirical practices in applied work. This approach has substantially changed the perception of evidence-based economics among economists, policymakers, and the general public.

The main goal of this approach is to evaluate (*ex-post*) the impact of an existing policy by comparing the distribution of a chosen outcome variable for individuals affected by the policy (the treatment group) with that of otherwise comparable individuals who are not affected (the control group). The central challenge is to construct this comparison in such a way that the outcomes of the control group provide a credible counterfactual for what would have happened to treated individuals in the absence of treatment. The focus is therefore on understanding and exploiting sources of variation in the data that allow the identification of causal policy parameters.

The main advantage of the treatment effects approach is that, by prioritizing internal validity, it delivers transparent and credible identification of causal effects within a well-defined setting. Its main limitation is that the estimated parameters are typically not suitable for welfare analysis, as they are not deep (structural) parameters but reduced-form causal effects, and therefore they are not policy-invariant. As a result, treatment effect estimates are generally informative about the impact of a specific policy in a specific environment, but less informative about

the effects of alternative policies or the same policy implemented in a different context.

In order to conduct a treatment effects analysis—or, more generally, any econometric analysis aimed at causal inference—we must address four fundamental questions:

- What is the causal relationship of interest?
- What ideal experiment would identify this causal effect?
- What is the empirical *identification strategy* (in the terminology of Angrist and Krueger, 1999) that allows us to approximate this experiment using observational data?
- What is the estimation and inference procedure? This includes the population of interest, the sample used, and the assumptions underlying standard errors and confidence intervals.

II. Potential Outcomes, Selection Bias, and Treatment Effects

Consider a population of individuals who are susceptible to a treatment. Let Y_{1i} denote the outcome for individual i if exposed to the treatment ($D_i = 1$), and let Y_{0i} denote the outcome for the same individual if not exposed to the treatment ($D_i = 0$). The ***individual treatment effect*** for individual i is defined as $Y_{1i} - Y_{0i}$. The variables Y_{1i} and Y_{0i} are referred to as ***potential outcomes***, reflecting the fact that at most one of them is observed for each individual:

$$Y_i = Y_{1i}D_i + Y_{0i}(1 - D_i). \tag{1}$$

Equation (1) highlights the fundamental problem of causal inference: the individual treatment effect cannot be observed, since we never observe both Y_{1i} and Y_{0i} for the same individual. Fortunately, our interest is typically not in individual-level treatment effects per se, but in features of their distribution, such as averages.

We focus primarily on two parameters of interest. The first is the ***average treatment effect*** (ATE):

$$\alpha_{ATE} \equiv \mathbb{E}[Y_{1i} - Y_{0i}], \tag{2}$$

which measures the average causal effect of the treatment in the population. The second is the ***average treatment effect on the treated*** (TT):

$$\alpha_{TT} \equiv \mathbb{E}[Y_{1i} - Y_{0i} \mid D_i = 1], \tag{3}$$

which measures the average effect of the treatment for individuals who receive it.

The distinction between these two parameters is subtle but important. The ATE is often the most relevant parameter for *ex-ante* policy design, but it is typically difficult to identify. By contrast, the TT is often easier to identify in practice, since it conditions on the treated population, for whom the treated outcome Y_{1i} is observed but not the untreated one Y_{0i}, which is counterfactual.

Let β denote the difference in mean observed outcomes between treated and untreated individuals:

$$\begin{aligned}\beta &\equiv \mathbb{E}[Y_i \mid D_i = 1] - \mathbb{E}[Y_i \mid D_i = 0] \\ &= \underbrace{\mathbb{E}[Y_{1i} - Y_{0i} \mid D_i = 1]}_{\alpha_{TT}} + \underbrace{\mathbb{E}[Y_{0i} \mid D_i = 1] - \mathbb{E}[Y_{0i} \mid D_i = 0]}_{\text{selection bias}}. \end{aligned} \tag{4}$$

The second term is referred to as ***selection bias***. It captures systematic differences in untreated potential outcomes between treated and untreated individuals:

$$\text{selection bias} \equiv \mathbb{E}[Y_{0i} \mid D_i = 1] - \mathbb{E}[Y_{0i} \mid D_i = 0]. \tag{5}$$

Selection bias arises when treatment status is related to individual characteristics that also affect outcomes. For example, in the hospital case, treated individuals (those who are hospitalized) would typically be less healthy even if they were sent home, implying a negative selection bias. As a result, a naive comparison of health outcomes between hospitalized and non-hospitalized individuals understates the causal effect of hospitalization.

An important assumption underlying the potential outcomes framework is the ***Stable Unit Treatment Value Assumption*** (SUTVA). SUTVA requires that an individual's potential outcomes depend only on her own treatment status, and not on the treatment received by other individuals, and that there are no hidden versions of the treatment. This rules out equilibrium effects, spillovers, and strategic interactions across units. As a consequence, the framework is best suited to the evaluation of policies whose effects are primarily individual-level rather than system-wide.

Sample analogs of the population parameters α_{ATE} and α_{TT} are given by:

$$\alpha^S_{ATE} \equiv \frac{1}{N}\sum_{i=1}^{N}(Y_{1i} - Y_{0i}), \tag{6}$$

$$\alpha^S_{TT} \equiv \frac{1}{N_1}\sum_{i=1}^{N} D_i(Y_{1i} - Y_{0i}), \tag{7}$$

where $N_1 \equiv \sum_{i=1}^{N} D_i$ denotes the number of treated individuals. If both factual and counterfactual potential outcomes were observed, these quantities could be estimated without error.

In practice, however, potential outcomes are not jointly observed. The sample analog of the naive difference in means estimator is therefore:

$$\begin{aligned} \beta^S &\equiv \bar{Y}_T - \bar{Y}_C \\ &\equiv \frac{1}{N_1}\sum_{i=1}^{N} Y_i D_i - \frac{1}{N_0}\sum_{i=1}^{N}(1-D_i)Y_i, \end{aligned} \tag{8}$$

where $N_0 \equiv N - N_1$ denotes the number of untreated (control) individuals.

Although the distinction between population and sample average treatment effects is not operationally important for identification, it matters for statistical inference. In particular, whether one treats the treatment effects as fixed attributes of the sample or as random draws from a population affects the construction of standard errors and confidence intervals.

III. Identification of Treatment Effects under Different Assumptions

The identification of treatment effects depends critically on the assumptions made about the relationship between treatment assignment and potential outcomes. Different assumptions correspond to different empirical strategies, which will be studied in detail in subsequent chapters.

The simplest case, ***independence***, arises when potential outcomes are independent of treatment assignment:

$$(Y_{1i}, Y_{0i}) \perp\!\!\!\perp D_i. \tag{9}$$

This assumption is most naturally satisfied in randomized experiments, where individuals are assigned to treatment or control purely at random. Under this assumption, the distributions of potential outcomes do not differ across treatment status: $F(Y_{1i} \mid D_i = 1) = F(Y_{1i})$ and $F(Y_{0i} \mid D_i = 0) = F(Y_{0i})$. As a result:

$$\mathbb{E}[Y_{1i}] = \mathbb{E}[Y_i \mid D_i = 1], \qquad \mathbb{E}[Y_{0i}] = \mathbb{E}[Y_i \mid D_i = 0], \tag{10}$$

and therefore $\alpha_{ATE} = \alpha_{TT} = \beta$. An unbiased estimator of the ATE is given by the difference in average outcomes between treated and control individuals:

$$\widehat{\alpha}_{ATE} = \bar{Y}_T - \bar{Y}_C = \beta^S. \tag{11}$$

In this context, there is no need to control for additional covariates in order to

identify the average treatment effect, unless one is interested in estimating heterogeneous effects across subpopulations or improving the precision of the estimator.

A less restrictive assumption is ***conditional independence***:

$$(Y_{1i}, Y_{0i}) \perp\!\!\!\perp D_i \mid X_i, \tag{12}$$

where X_i is a vector of observed covariates. This assumption states that, conditional on X_i, treatment assignment is as good as random. Conditional independence implies that, for any value of X_i:

$$\mathbb{E}[Y_{1i} \mid X_i] = \mathbb{E}[Y_i \mid D_i = 1, X_i], \qquad \mathbb{E}[Y_{0i} \mid X_i] = \mathbb{E}[Y_i \mid D_i = 0, X_i]. \tag{13}$$

As a result, the average treatment effect can be written as:

$$\begin{aligned} \alpha_{ATE} &= \mathbb{E}[Y_{1i} - Y_{0i}] \\ &= \mathbb{E}[\mathbb{E}[Y_{1i} - Y_{0i} \mid X_i]] \\ &= \int (\mathbb{E}[Y_i \mid D_i = 1, X_i] - \mathbb{E}[Y_i \mid D_i = 0, X_i])\, dF(X_i). \end{aligned} \tag{14}$$

Similarly, the average treatment effect on the treated can be expressed as:

$$\begin{aligned} \alpha_{TT} &= \int \mathbb{E}[Y_{1i} - Y_{0i} \mid D_i = 1, X_i]\, dF(X_i \mid D_i = 1) \\ &= \int \mathbb{E}[Y_i - \mu_0(X_i) \mid D_i = 1, X_i]\, dF(X_i \mid D_i = 1), \end{aligned} \tag{15}$$

where $\mu_0(X_i) \equiv \mathbb{E}[Y_i \mid D_i = 0, X_i]$ is used as an imputation rule for the missing counterfactual outcome Y_{0i} of treated individuals.

These estimands describe what is known as matching: for each "type" of individual (i.e. each value of covariates) we can match treated and control individuals, so that the latter act as counterfactuals for the former. It is important to emphasize, however, that conditional independence is an identifying assumption rather than a mechanical fix. Its plausibility depends on whether the covariates X_i adequately capture all factors jointly affecting treatment assignment and potential outcomes.

In many empirical settings, conditional independence is implausible:

$$(Y_{1i}, Y_{0i}) \not\perp\!\!\!\perp D_i \mid X_i. \tag{16}$$

In such cases, identification may still be possible if there exists an additional variable Z_i (an ***instrumental variable***) that provides an exogenous source of variation in treatment. Specifically, the variable Z_i must satisfy two conditions. First, it must be independent of potential outcomes conditional on X_i:

$$(Y_{1i}, Y_{0i}) \perp\!\!\!\perp Z_i \mid X_i. \tag{17}$$

Second, it must be relevant for treatment assignment:

$$Z_i \not\perp\!\!\!\perp D_i \mid X_i. \tag{18}$$

In addition, identification requires that Z_i affects the outcome only through its effect on treatment (an exclusion restriction). As discussed in Chapter 4, under these assumptions we generally identify a *local average treatment effect* for a subpopulation of individuals.

IV. Linear Regression and Treatment Effects

A. *Formal connection between regression and treatment effects*

The potential outcomes framework provides a clear language for thinking about causality, but it can be cumbersome in practice. It is therefore useful to understand how treatment effects relate to linear regression models commonly used in applied work. Rearranging Equation (1) yields:

$$\begin{aligned} Y_i &= Y_{0i}(1 - D_i) + Y_{1i}D_i \\ &= Y_{0i} + (Y_{1i} - Y_{0i})D_i \\ &= \mathbb{E}[Y_{0i}] + (Y_{1i} - Y_{0i})D_i + (Y_{0i} - \mathbb{E}[Y_{0i}]) \\ &\equiv \beta_0 + \beta_i D_i + U_i. \end{aligned} \tag{19}$$

Equation (19) expresses the outcome as a linear function of the treatment indicator, where $\beta_i \equiv Y_{1i} - Y_{0i}$ is the individual-specific (possibly heterogeneous) causal effect of the treatment, and $U_i \equiv Y_{0i} - \mathbb{E}[Y_{0i}]$ is the regression error term.

To fix ideas, suppose first that treatment effects are homogeneous, so that $\beta_i = \beta$ for all individuals. In this case, $Y_{1i} - Y_{0i}$ is constant, and the causal effect of treatment is equal to both the average treatment effect and the treatment effect on the treated: $\alpha_{ATE} = \alpha_{TT} = \beta$. A linear regression of Y_i on D_i consistently estimates β provided that the error term U_i, or equivalently Y_{0i}, is mean-independent of the treatment indicator D_i.

When treatment effects are heterogeneous, a linear regression of Y_i on D_i no longer recovers the ATE in general. Consider instead the linear regression:

$$Y_i = \beta_0 + \bar{\beta} D_i + U_i. \tag{20}$$

The slope coefficient $\bar{\beta}$ summarizes the average relationship between the outcome and treatment in the data. Recall that in a linear regression with an intercept,

the slope coefficient can be written as:

$$\bar{\beta} = \frac{\mathrm{Cov}(Y_i, D_i)}{\mathrm{Var}(D_i)} = \frac{\mathbb{E}[Y_i D_i] - \mathbb{E}[Y_i]\,\mathbb{E}[D_i]}{\mathbb{E}[D_i^2] - \mathbb{E}[D_i]^2}. \tag{21}$$

Since D_i is binary, $D_i^2 = D_i$, and therefore:

$$\mathrm{Var}(D_i) = \mathbb{E}[D_i^2] - \mathbb{E}[D_i]^2 = \mathbb{E}[D_i](1 - \mathbb{E}[D_i]). \tag{22}$$

To evaluate the numerator, we use the law of iterated expectations and note that $\Pr(D_i = 1) = \mathbb{E}[D_i]$. We obtain:

$$\begin{aligned}\mathbb{E}[Y_i D_i] &= \mathbb{E}[Y_i \cdot 1 | D_i = 1]\,\Pr(D_i = 1) + \mathbb{E}[Y_i \cdot 0 | D_i = 0]\,\Pr(D_i = 0) \\ &= (\beta_0 + \mathbb{E}[\beta_i \mid D_i = 1] + \mathbb{E}[U_i \mid D_i = 1])\,\mathbb{E}[D_i]. \end{aligned} \tag{23}$$

Similarly, the unconditional expectation of Y_i can be written as:

$$\begin{aligned}\mathbb{E}[Y_i] &= \mathbb{E}[Y_i \mid D_i = 1]\,\mathbb{E}[D_i] + \mathbb{E}[Y_i \mid D_i = 0](1 - \mathbb{E}[D_i]) \\ &= \beta_0 + \mathbb{E}[\beta_i \mid D_i = 1]\,\mathbb{E}[D_i] + \mathbb{E}[U_i \mid D_i = 1]\,\mathbb{E}[D_i] + \mathbb{E}[U_i \mid D_i = 0](1 - \mathbb{E}[D_i]). \end{aligned} \tag{24}$$

Combining these expressions, the covariance between Y_i and D_i is:

$$\begin{aligned}\mathrm{Cov}(Y_i, D_i) &= \mathbb{E}[Y_i D_i] - \mathbb{E}[Y_i]\,\mathbb{E}[D_i] \\ &= \{\mathbb{E}[\beta_i \mid D_i = 1] + (\mathbb{E}[U_i \mid D_i = 1] - \mathbb{E}[U_i \mid D_i = 0])\}\,\mathbb{E}[D_i](1 - \mathbb{E}[D_i]). \end{aligned} \tag{25}$$

Dividing by $\mathrm{Var}(D_i)$ yields:

$$\bar{\beta} = \mathbb{E}[Y_{1i} - Y_{0i} \mid D_i = 1] + (\mathbb{E}[Y_{0i} \mid D_i = 1] - \mathbb{E}[Y_{0i} \mid D_i = 0])\,. \tag{26}$$

Thus, in the presence of heterogeneous treatment effects, the OLS coefficient identifies the average treatment effect on the treated plus selection bias.

B. Selection bias in regression: education example

We illustrate the connection between regression analysis and treatment effects using a classic example: the effect of education on wages. Let C_i denote the treatment indicator, where $C_i = 1$ if individual i attends college and $C_i = 0$ otherwise. Let Y_{1i} denote the earnings of individual i if she attends college, and Y_{0i} her earnings if she does not.

A regression of observed wages on a college dummy estimates the coefficient discussed in the previous section. In this context, however, selection bias is likely to be present. Individuals with higher ability —broadly interpreted to include

both cognitive and non-cognitive skills— are more likely to attend college and also tend to earn higher wages regardless of their education level. As a result, individuals with $C_i = 1$ have, on average, higher potential earnings than those with $C_i = 0$, even in the absence of college education.

Formally, this implies that:

$$\mathbb{E}[Y_{0i} \mid C_i = 1] - \mathbb{E}[Y_{0i} \mid C_i = 0] > 0, \tag{27}$$

that is, the regression coefficient overstates the causal effect of college attendance: $\bar{\beta} > \alpha_{TT}$. If we are able to control for ability, denoted by A_i, the conditional independence assumption becomes more plausible:

$$\mathbb{E}[Y_{0i} \mid A_i, C_i = 1] - \mathbb{E}[Y_{0i} \mid A_i, C_i = 0] = 0. \tag{28}$$

In words, conditional on ability, potential earnings without college of individuals who attend college are not systematically different from those who do not.

C. Omitted variable bias

In regression terms, controlling for ability amounts to estimating:

$$Y_i = \beta_0 + \beta C_i + \gamma A_i + U_i, \tag{29}$$

possibly augmented with interactions between C_i and A_i. Under the conditional independence assumption, this "long" regression has a causal interpretation, whereas the "short" regression that omits A_i does not.

The ***omitted variable bias formula*** formalizes the relationship between the coefficients in these two regressions. Define $\tilde{U}_i \equiv \gamma A_i + U_i$, and rewrite Equation (29) as:

$$Y_i = \beta_0 + \beta C_i + \tilde{U}_i. \tag{30}$$

The slope coefficient from this regression, denoted by $\tilde{\beta}$, is:

$$\begin{aligned} \tilde{\beta} &= \frac{\mathrm{Cov}(Y_i, C_i)}{\mathrm{Var}(C_i)} \\ &= \beta + \frac{\mathrm{Cov}(\tilde{U}_i, C_i)}{\mathrm{Var}(C_i)}. \end{aligned} \tag{31}$$

The second term captures the bias induced by omitting A_i from the regression.

The omitted variable bias can be expressed as:

$$\begin{aligned}\frac{\mathrm{Cov}(\tilde{U}_i, C_i)}{\mathrm{Var}(C_i)} &= \frac{\mathrm{Cov}(\gamma A_i + U_i, C_i)}{\mathrm{Var}(C_i)} \\ &= \gamma \frac{\mathrm{Cov}(A_i, C_i)}{\mathrm{Var}(C_i)}. \end{aligned} \tag{32}$$

The covariance between U_i and C_i is equal to zero under the conditional independence assumption. Specifically,

$$\begin{aligned}\mathrm{Cov}(U_i, C_i) &= \mathbb{E}[U_i C_i] - \mathbb{E}[U_i]\,\mathbb{E}[C_i] \\ &= \mathbb{E}[\mathbb{E}[U_i \mid A_i, C_i] C_i] = 0, \end{aligned} \tag{33}$$

since $\mathbb{E}[U_i \mid A_i, C_i] = 0$ by construction. Thus, omitted variable bias depends on two components: the effect of the omitted variable on the outcome (γ), and the association between the omitted variable and the treatment. In the schooling example, ability is positively correlated with both schooling and wages, implying a positive omitted variable bias and an overestimation of the returns to schooling.

To illustrate these ideas empirically, we reproduce below Table 3.2.1 from Angrist and Pischke, which reports estimates of the return to schooling using data from the National Longitudinal Survey of Youth (1979 cohort). A key advantage of this dataset is the availability of the Armed Forces Qualification Test (AFQT) score, which serves as a proxy for ability.

	(1)	(2)	(3)	(4)	(5)
Controls:	None	Age Dummies	Col. (2) and Additional Controls*	Col. (3) and AFQT score	Col. (2) with Occupation Dummies
	0.132	0.131	0.114	0.087	0.066
	(0.007)	(0.007)	(0.007)	(0.009)	(0.010)

Note: Data are from the National Longitudinal Survey of Youth (1979 cohort, 2002 survey). The table reports the coefficient on years of schooling in a regression of log wages on years of schooling and the indicated controls. Standard errors are shown in parentheses. The sample is restricted to men and weighted by NLSY sampling weights. The sample size is 2,434.
* Additional controls are mother's and father's education and dummies for race and census region.

The results suggest that ability is an important omitted variable. The coefficient on schooling declines when additional background controls are included and falls substantially once AFQT is added. This pattern is consistent with ability being positively correlated with both schooling and wages, generating a positive omitted variable bias in simpler specifications.

Two additional remarks are worth emphasizing. First, years of schooling—the treatment variable in this example—is not binary, but takes multiple values. Sec-

ond, the interpretation of the specification including occupation dummies is not straightforward. The remainder of this chapter addresses these two issues in turn.

D. Treatment variables that take more than two values

The schooling example motivates a discussion of treatments that take more than two discrete values, or that are naturally viewed as continuous. In this case, we define a treatment effect function:

$$Y_{si} \equiv f_i(s), \tag{34}$$

where $f_i(s)$ represents the potential outcome for individual i if the treatment intensity were equal to s. In the schooling example, $Y_{1i} = f_i(16)$ and $Y_{0i} = f_i(12)$ correspond to college and high school completion, respectively. More generally, the function $f_i(s)$ answers a causal "what if" question for every possible value of schooling s, even if the observed schooling level S_i takes a particular value.

E. Endogenous (or "bad") controls

We conclude by revisiting the inclusion of occupation dummies in the schooling regression. Occupation is a *post-treatment* (or "bad") control, since educational attainment affects occupational choice.

Let the observed white-collar indicator be defined as:

$$W_i = W_{1i}C_i + W_{0i}(1 - C_i), \tag{35}$$

where $W_i = \mathbb{1}\{\text{white collar}\}$. Even if college attendance were randomly assigned, education would causally affect occupational choice:

$$\mathbb{E}[W_i \mid C_i = 1] - \mathbb{E}[W_i \mid C_i = 0] = \mathbb{E}[W_{1i} - W_{0i}]. \tag{36}$$

If we instead estimate the effect of college on wages conditional on occupation, for example among white-collar workers, we obtain:

$$\mathbb{E}[Y_i \mid W_i = 1, C_i = 1] - \mathbb{E}[Y_i \mid W_i = 1, C_i = 0]. \tag{37}$$

This comparison does not have a clear causal interpretation.

Indeed, this expression can be decomposed as:

$$\begin{aligned} &\mathbb{E}[Y_{1i} \mid W_{1i} = 1] - \mathbb{E}[Y_{0i} \mid W_{0i} = 1] \\ &= \underbrace{\mathbb{E}[Y_{1i} - Y_{0i} \mid W_{1i} = 1]}_{\text{causal effect}} + \underbrace{\mathbb{E}[Y_{0i} \mid W_{1i} = 1] - \mathbb{E}[Y_{0i} \mid W_{0i} = 1]}_{\text{selection bias}}. \end{aligned} \tag{38}$$

The decline in the estimated return to schooling when occupation dummies are included is therefore difficult to interpret. The induced selection bias may be positive or negative, depending on how occupational choice, education, and potential earnings are related. As a result, controlling for post-treatment variables typically answers a different—and often policy-irrelevant—causal question. In practice, one should restrict attention to controls that are not themselves affected by the treatment.